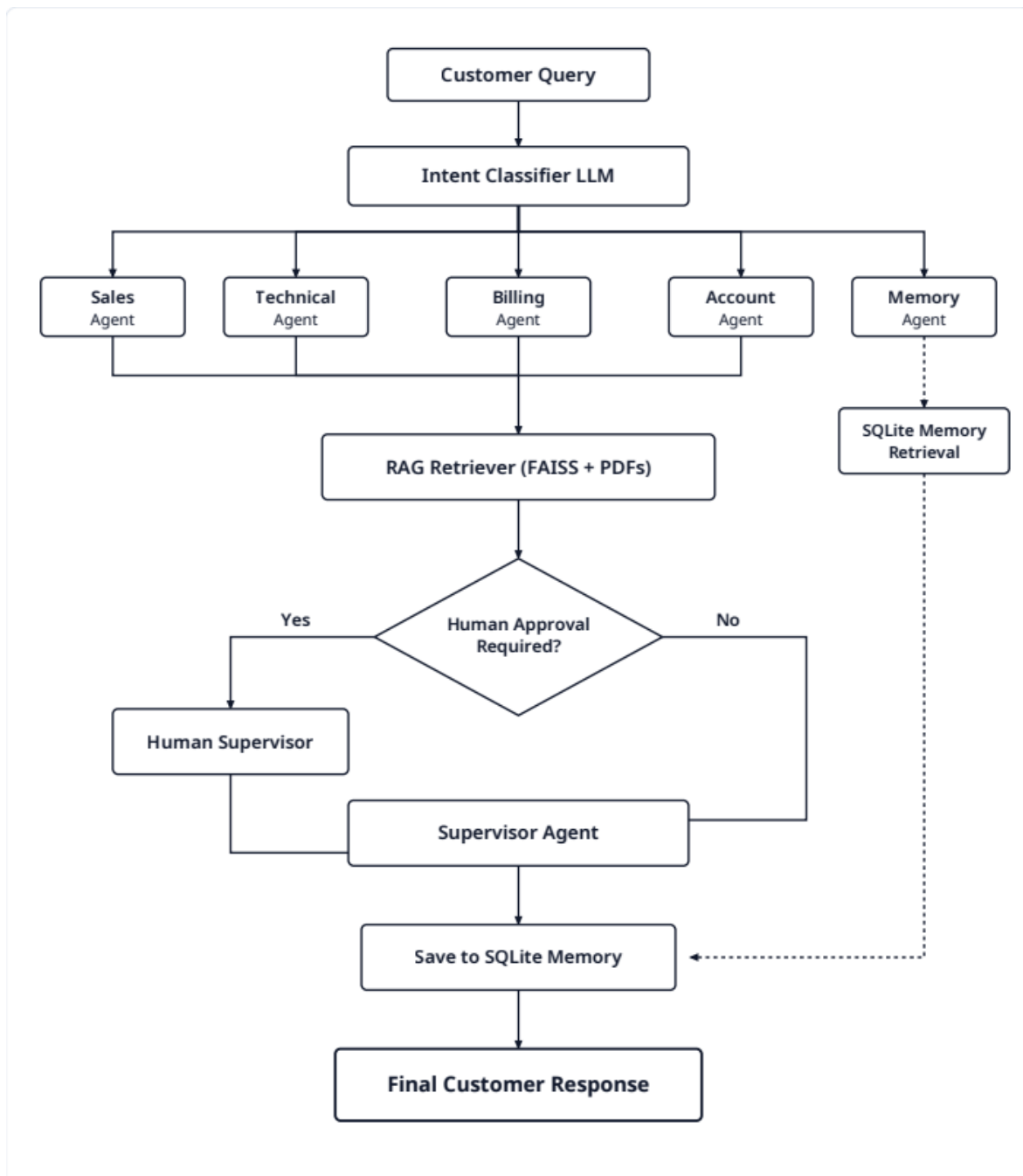


IBM AGENTIC AI

ASSIGNMENT 2

ABC Technologies Customer Support Automation System Using LangGraph

System Architecture:



1. Customer enters a query.
2. The Intent Classifier determines the query category.
3. LangGraph routes the request to the appropriate agent.
4. The selected agent retrieves relevant information using the RAG pipeline.
5. Sensitive requests are sent for Human Approval.
6. The Supervisor Agent reviews the response.
7. The interaction is stored in SQLite memory.
8. The final response is returned to the customer.

The Intent Classification Agent is responsible for understanding customer queries and identifying the appropriate department.

The project contains four department-specific agents.

Sales Agent

Handles:

- Product information
- Pricing
- Subscription plans

Technical Agent

Handles:

- Login issues
- Installation
- Configuration
- Application errors

Billing Agent

Handles:

- Payments
- Refunds
- Invoices

Account Agent

Handles:

- Password reset
- Profile updates

- Account activation
- Account deactivation

The project uses Retrieval-Augmented Generation (RAG) to answer customer queries using company documents.

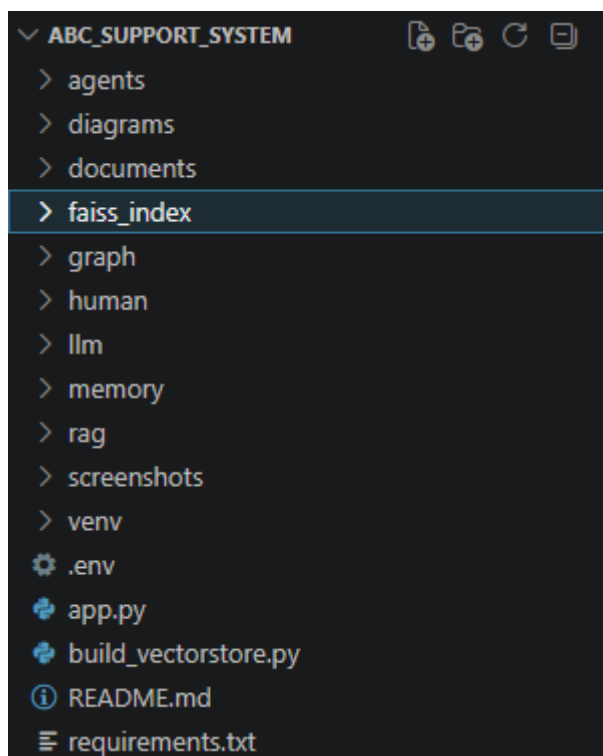
SQLite is used to maintain customer conversation history.

Certain requests that cannot be automatically approved are forwarded to a human supervisor before the final response is generated.

The Supervisor Agent performs the final quality check.

Demonstration:

Project Structure:



Agent Routing:

```
(venv) PS D:\Projects\abc_support_system> python app.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|██████████| 103/103 [00:00<00:00, 1280.73it/s]
=====
ABC TECHNOLOGIES CUSTOMER SUPPORT BOT
=====

Enter your name: Nisha

Customer: What are the pricing plans available for your software?

Bot:

We have three subscription plans available:
1. Starter Plan: ₹499 per month
2. Professional Plan: ₹999 per month
3. Enterprise Plan: Custom Pricing

A 14-day free trial is also available, with no credit card required.

Customer: I forgot my account password.

Bot:

To reset your password, please click on the "Forgot Password" option on the login page and follow the instructions. You will receive a password reset email to help you regain access to your account. If you need further assistance, you can contact our support team through live chat, email, or our support portal.

Customer: My application crashes whenever I upload a file

Bot:

To resolve the issue with your application crashing while uploading files, please try the following steps:

1. Refresh the application.
2. Clear browser cache.
3. Check your internet connection.
4. Try using another browser.

If the issue continues, please contact Technical Support for further assistance. Also, ensure the file meets the upload requirements:
- Maximum upload size: 100 MB
- Supported file types: PDF, DOCX, XLSX, PNG, JPG. Large files should be compressed before uploading.

Customer: I need a refund for my annual subscription
=====
```

HUMAN SUPERVISOR APPROVAL REQUIRED

Customer Query:

I need a refund for my annual subscription

AI Draft Response:

I'd be happy to help you with your refund request. Since you're requesting a refund for your annual subscription, I'll need to inform you that refund requests are subject to review and approval by a human supervisor.

To proceed, I'll need to submit your refund request for review. Please note that refund requests are not automatically approved, and every request must be reviewed and approved by a supervisor before processing. If approved, refunds are typically processed within 5-7 business days.

Also, keep in mind that you are eligible for a refund review since you are requesting it within 30 days of purchasing your subscription.

I'll go ahead and submit your refund request for review. You will be notified once a decision has been made. If you have any further questions or concerns, please don't hesitate to reach out to us through email, live chat, or our support portal.

Approve this response? (yes/no): yes

Bot:

I'd be happy to help you with your refund request. Since you're requesting a refund for your annual subscription, I'll need to inform you that refund requests are subject to review and approval.

To proceed, I'll submit your refund request for review. Refunds are typically processed within 5-7 business days if approved. You are eligible for a refund review since you are requesting it within 30 days of purchasing your subscription.

I'll notify you once a decision has been made. If you have any further questions or concerns, please don't hesitate to reach out to us.

Customer: what was my previous support issue?

Bot:

Your previous support issue was regarding a refund for your annual subscription.

Customer: exit

Thank you for using ABC Technologies Support.

RAG Retrieval

Customer: What are the pricing plans available for your software?

Bot:

We have three subscription plans available:

1. Starter Plan: ₹499 per month
2. Professional Plan: ₹999 per month
3. Enterprise Plan: Custom Pricing

A 14-day free trial is also available, with no credit card required.

Human Approval:

Approve this response? (yes/no): yes

Bot:

I'd be happy to help you with your refund request. Since you're requesting a refund for your annual subscription, I'll need to inform you that refund requests are subject to review and approval.

To proceed, I'll submit your refund request for review. Refunds are typically processed within 5-7 business days if approved. You are eligible for a refund review since you are requesting it within 30 days of purchasing your subscription.

I'll notify you once a decision has been made. If you have any further questions or concerns, please don't hesitate to reach out to us.

Approve this response? (yes/no): no

Bot:

Your request for a refund of your annual subscription has been forwarded for review. You will receive an update shortly.

Final Response:

Customer: I need a refund for my annual subscription

=====

HUMAN SUPERVISOR APPROVAL REQUIRED

=====

Customer Query:

I need a refund for my annual subscription

AI Draft Response:

I'd be happy to help you with your refund request. Since you're requesting a refund for your annual subscription, I'll need to inform you that refund requests are subject to review and approval by a human supervisor.

To proceed, I'll need to submit your refund request for review. Please note that refund requests are not automatically approved, and every request must be reviewed and approved by a supervisor before processing. If approved, refunds are typically processed within 5-7 business days.

Also, keep in mind that you are eligible for a refund review since you are requesting it within 30 days of purchasing your subscription.

I'll go ahead and submit your refund request for review. You will be notified once a decision has been made. If you have any further questions or concerns, please don't hesitate to reach out to us through email, live chat, or our support portal.

Approve this response? (yes/no): yes

Bot:

I'd be happy to help you with your refund request. Since you're requesting a refund for your annual subscription, I'll need to inform you that refund requests are subject to review and approval.

To proceed, I'll submit your refund request for review. Refunds are typically processed within 5-7 business days if approved. You are eligible for a refund review since you are requesting it within 30 days of purchasing your subscription.

I'll notify you once a decision has been made. If you have any further questions or concerns, please don't hesitate to reach out to us.

Memory Recall:

```
Customer Query:
I need a refund for my annual subscription

AI Draft Response:

I'd be happy to help you with your refund request. Since you're requesting a refund for your annual subscription, I'll need to inform you that refund requests are subject to review and approval by a human supervisor.

To proceed, I'll need to submit your refund request for review. Please note that refund requests are not automatically approved, and every request must be reviewed and approved by a supervisor before processing. If approved, refunds are typically processed within 5-7 business days.

Also, keep in mind that you are eligible for a refund review since you are requesting it within 30 days of purchasing your subscription.

I'll go ahead and submit your refund request for review. You will be notified once a decision has been made. If you have any further questions or concerns, please don't hesitate to reach out to us through email, live chat, or our support portal.

Approve this response? (yes/no): yes

Bot:

I'd be happy to help you with your refund request. Since you're requesting a refund for your annual subscription, I'll need to inform you that refund requests are subject to review and approval.

To proceed, I'll submit your refund request for review. Refunds are typically processed within 5-7 business days if approved. You are eligible for a refund review since you are requesting it within 30 days of purchasing your subscription.

I'll notify you once a decision has been made. If you have any further questions or concerns, please don't hesitate to reach out to us.

Customer: what was my previous support issue?

Bot:

Your previous support issue was regarding a refund for your annual subscription.
```

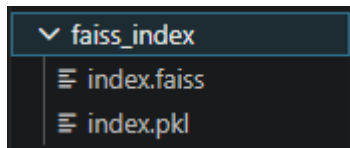
SQLite Database:

memory > memory.db

Rows: 10

id	customer...	query	response
1	David	I have a billing issue.	I'd be happy to help you with your billing issue. Could y...
2	Nisha	What are the pricing plans available for your software?	We have three subscription plans available: 1. Starter Pla...
3	Nisha	I forgot my account password	To reset your password, please click on the "Forgot Pass...
4	Nisha	My application crashes whenever I upload a file.	To resolve the issue with your application crashing while...
5	Nisha	I need a refund for my annual subscription	Your request has been forwarded to our team for review...
6	Nisha	What are the pricing plans available for your software?	We have three subscription plans available: 1. Starter Pla...
7	Nisha	I forgot my account password.	To reset your password, please click on the "Forgot Pass...
8	Nisha	My application crashes whenever I upload a file	To resolve the issue with your application crashing while...
9	Nisha	I need a refund for my annual subscription	I'd be happy to help you with your refund request. Since...
10	Nisha	I need a refund for my annual subscription	Your request for a refund of your annual subscription ha...
11			

FAISS Index:



Folder Structure:

agents/

Contains all specialized AI agents.

- classifier.py
- sales.py
- technical.py
- billing.py
- account.py
- supervisor.py
- memory_agent.py

graph/

Contains LangGraph workflow and state definitions.

rag/

Implements Retrieval-Augmented Generation.

memory/

Maintains SQLite conversation history.

human/

Contains Human-in-the-Loop approval implementation.

Github repository link: